

YASSER AHMED MOGAHED

Data Scientist | Data Analyst | BI & ML Engineer | Team Leader

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PROFESSIONAL SUMMARY

Results-driven Data Scientist and Data Analyst with 2+ years of project experience designing end-to-end machine learning pipelines, statistical models, NLP classifiers, and production-grade business intelligence platforms. Proficient in Python, SQL, Power BI, FastAPI, Azure, Docker, and MLflow. Skilled in data wrangling, predictive modeling, data visualization, and translating complex datasets into actionable, data-driven insights for executive stakeholders. Proven team leader experienced in agile project management, cross-functional collaboration, A/B testing, model interpretability, and stakeholder reporting — committed to delivering measurable business impact through data.

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Streamlit, Flask), R, SQL (Joins, Window Functions, CTEs, Stored Procedures), C++, JavaScript (ES6+), HTML5, CSS3

ML & Statistics: Supervised Learning, Regression, Classification, Deep Learning (fundamentals), NLP (TF-IDF, Sentiment Analysis, Text Preprocessing), K-Means Clustering, RFM Analysis, PCA, ROC/AUC, A/B Testing, Statistical Hypothesis Testing, Hyperparameter Tuning (Optuna, GridSearchCV), Feature Engineering, Feature Selection, Cross-Validation, Model Deployment, Predictive Analytics, Data-Driven Decision Making

BI & Visualization: Power BI (DAX, Data Modeling, Star Schema, Interactive Dashboards, Drill-Through, Slicers, Report Design), Excel (Pivot Tables, Power Query, VLOOKUP, Advanced Formulas), Matplotlib, Seaborn, Plotly, Tableau (fundamentals)

Engineering & Tools: ETL Pipelines, Data Pipeline Development, Exploratory Data Analysis (EDA), Data Cleaning, Data Wrangling, Statistical Analysis, Medallion Architecture, T-SQL Stored Procedures, REST API Development (FastAPI), Full-Stack Web Development (HTML/CSS/JS), Docker Containerization, MLflow Experiment Tracking, Azure Cloud Computing (SQL, Blob Storage, Deployment), Git, GitHub Actions (CI/CD), Agile Project Management, Jupyter Notebook, VS Code

SOFT SKILLS

Analytical Thinking & Problem Solving • Data Storytelling & Communication • Team Leadership & Stakeholder Management • Attention to Detail & Critical Thinking • Time Management & Results-Oriented • Adaptability & Continuous Learning

WORK EXPERIENCE

Data Science Member — Internship | Microsoft Student Club | Kafr El Sheikh University | Feb 2026 – May 2026

- Participated in 10+ structured training sessions and 15+ hands-on workshops covering end-to-end machine learning pipeline development in Python, Pandas, NumPy, and Scikit-learn — demonstrating measurable proficiency in EDA, feature engineering, and supervised model evaluation (Accuracy, RMSE, ROC/AUC) within 4 months.
- Cleaned, transformed, and analyzed 3+ real-world datasets using Pandas and NumPy; engineered predictive features and developed classification and regression pipelines with k-fold cross-validation — improving model accuracy and reinforcing clean-code standards through 5+ structured peer code reviews.

PROJECTS

Car Resale Price Prediction & Market Intelligence Dashboard | Python, Scikit-learn, Random Forest, ElasticNet, Streamlit | 2025 | https://github.com/NitoBoritto/Car_Sales_Regression_Analysis

- Led a 5-person team to design and deliver a production-ready, end-to-end vehicle price prediction system trained on 16,733 real-world car listings; applied feature engineering (10+ features), Box-Cox transformation, and hyperparameter tuning — achieving R^2 of 0.8244 with Random Forest vs. 0.6724 ElasticNet baseline (+22.6% improvement).
- Deployed a 3-page interactive Streamlit dashboard featuring KPI cards, brand deep-dive visualizations, and a real-time AI Price Predictor with MAE-based confidence ranges — enabling instant valuation for any vehicle in the dataset.

ERU Timetable Scheduler — AI Capstone | Python, Streamlit, CSP, Backtracking Algorithm | 2025–2026 | github.com/Yasser-Mogahed

- Designed and implemented a Constraint Satisfaction Problem (CSP) solver with backtracking search, generating 100% conflict-free academic timetables for 50+ courses across 4 levels — enforcing 10+ hard constraints (room conflicts, instructor double-booking, mandatory breaks) with zero scheduling violations.
- Delivered a PDF export module in ERU's official format for 8+ specialization groups, eliminating 100% of manual scheduling effort and saving an estimated 3+ hours per scheduling cycle.

Maven Market BI Report — Multi-National Retail Chain | Power BI, DAX, Data Modeling, Excel | 2026 | github.com/Yasser-Mogahed

- Designed and engineered a 3-page Power BI KPI dashboard with 8+ custom DAX measures, drill-through navigation, and dynamic slicers — delivering sales trend analysis, customer segmentation insights, and regional performance tracking across 5+ countries and 3 product categories.
- Identified the 3 highest-impact root causes of missed revenue targets through drill-through analysis and data-driven recommendations — reducing ad-hoc reporting requests by 50%, saving ~10 analyst hours per month, and enabling full self-service analytics for C-suite stakeholders.

Olist E-Commerce BI & ML Platform | Python, FastAPI, Azure SQL, Power BI, MLflow, Docker, T-SQL | 2025–2026 |

https://github.com/NitoBoritto/Olist_Data_Department

- Led a 7-member cross-functional team to architect and deliver a production-grade, end-to-end data analytics and machine learning platform on the Brazilian Olist e-commerce dataset (~100K orders, 9 relational tables); directed data architecture design, NLP sentiment classification pipeline (Logistic Regression + TF-IDF, Optuna-tuned, MLflow-tracked), and K-Means RFM customer segmentation — producing 5 actionable customer personas that informed targeted business strategy.
- Sole developer of a full-stack web analytics platform (4 interactive dashboards, 3D K-Means scatter, geospatial customer map) in HTML/CSS/JS; deployed a real-time FastAPI sentiment inference REST API with Docker containerization and Azure cloud hosting — delivering sub-second prediction response time.

Bank Analytics — Datathon Competition (Data Science Track) | Python, HTML, CSS, JavaScript | 2024 |

https://github.com/NitoBoritto/El_Farghaly_Bros

- Led a 4-member team to 1st place in a college-level Bank Analytics Datathon competition; served as team leader and sole full-stack developer — independently designed and built the entire competition website from scratch using HTML, CSS, and JavaScript, delivering a polished, fully functional data analytics platform under competition conditions.
- Conducted end-to-end banking data analysis including customer segmentation, churn prediction, and KPI visualization; communicated findings through an interactive web interface — demonstrating applied data science skills in a high-pressure, real-world competitive environment, earning 1st place out of all competing teams.

Microsoft Student Club — ML Graduation Project | Python, Scikit-learn, Pandas, NumPy | 2026 | [https://github.com/Yasser-](https://github.com/Yasser-Mogahed/MS-ML-Project)

[Mogahed/MS-ML-Project](https://github.com/Yasser-Mogahed/MS-ML-Project)

- Independently designed and delivered a complete end-to-end machine learning project as the solo capstone for the Microsoft Student Club (MSC) Data Science program — officially graduated with a perfect mark, demonstrating mastery of the full ML pipeline from data preprocessing and feature engineering through model training, evaluation, and deployment-ready output.
- Applied supervised learning techniques including classification, regression, and model evaluation (Accuracy, RMSE, ROC/AUC); implemented structured data preprocessing workflows and cross-validation strategies — validated through rigorous MSC program assessment, receiving a perfect score on all evaluated criteria.

Task Manager — Data Structures Project | C++, Data Structures, UI Design | 2024 | <https://github.com/NitoBoritto/C---Task-Manager>

- Led a 6-member team to design and implement a fully functional Task Manager application in C++, applying core data structures concepts (linked lists, queues, stacks, trees) to manage task creation, prioritization, and scheduling — serving as team leader responsible for coordinating development milestones, code integration, and peer task assignments.
- Sole designer and developer of the application's user interface — built a clean, intuitive console-based UI enabling task CRUD operations, priority filtering, and real-time status updates; reinforced algorithmic problem-solving and software engineering best practices in a collaborative, version-controlled team environment.

EDUCATION

Bachelor of Science in Business Analytics | Egyptian Russian University (ERU), Cairo, Egypt | GPA: 3.00 | July 2027

Specialization: Business Technology | Relevant Coursework: Machine Learning, Predictive Analytics, Operations Research, Database Systems, Artificial Intelligence, Business Statistics, Data Structures & Algorithms, Computer Networks

CERTIFICATIONS

- Data Analysis Diploma — AMIT Learning (Oct 2025 – Feb 2026)

LANGUAGES

- Arabic — Native proficiency
- English — C1 Advanced proficiency
- French — Basic proficiency
- Russian — Basic proficiency